MCA03 – Due: Fri., Oct. 30, 2026, before 11:50 PM

Electrical and Computer Engineering Department
University of Puerto Rico at Mayagüez



COMMUNICATIONS THEORY I
MATLAB CLASS ASSIGNMENT 03 (MCA03)
Midrise Uniform Quantizer Simulations & Signal Performance Metrics

By

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INTRODUCTION

Signal model and notation

Let $x[n] \in \mathbb{R}$, $n = 0,1,\ldots,L-1$, denote a finite-length audio signal. Here L is the *number of samples*, F_s the sampling frequency (Hz), $T_s = 1/F_s$ the sampling interval, and $t_n = nT_s$ the sampling instants. Let $N = 2^B$ denote the *number of quantization levels*, where B is the number of bits per sample. These quantities must not be confused.

In the associated experiment, $L = 2205$ samples from dtmfad.wav are processed at $N \in \{4,8,16,32,64,128\}$ ($B = 2,\ldots,7$). At $F_s = 44100$ Hz, these samples represent $\frac{L}{F_s} = \mathrm{L} \cdot T_s = 0.05$ seconds. The definitions below apply independently to the original signal x and any quantized signal $x_{q,N}$.

Six fundamental signal metrics

The following finite-vector conventions follow the signal-metrics discussion in Smith's Book (see reference below) . Energy and average power here are *discrete-time, amplitude-squared metrics*; no sample-period or physical impedance factor is included.

1. **Arithmetic mean (DC component):**

$$\mu_x = \frac{1}{L}\sum_{n=0}^{L-1} x\,[n].$$

The mean measures the average signed amplitude; for audio, a nonzero mean indicates a DC offset. mean(x).

2. **Total signal energy:**

$$\mathcal{E}_x = \sum_{n=0}^{L-1} |x[n]|^2.$$

This is the squared Euclidean length of the signal vector. sum(abs(x).^2).

3. **Average signal power (mean square):**

$$\mathcal{P}_x = \frac{\mathcal{E}_x}{L} = \frac{1}{L}\sum_{n=0}^{L-1} |x[n]|^2.$$

This quantity is mean-square amplitude, not necessarily physical watts. mean(abs(x).^2).

4. **Root-mean-square (RMS) amplitude:**

$$x_{\mathrm{RMS}} = \sqrt{\mathcal{P}_x} = \sqrt{\frac{1}{L}\sum_{n=0}^{L-1} |x[n]|^2}.$$

RMS here includes any DC component. sqrt(mean(abs(x).^2)).

5. **Variance with population (L) normalization:**

$$\sigma_x^2 = \frac{1}{L}\sum_{n=0}^{L-1}|x[n]-\mu_x|^2 = \mathcal{P}_x - |\mu_x|^2.$$

Variance measures the mean-square amplitude after DC removal. var(x,1). MATLAB's var(x) normally uses $L-1$, which is *not* the convention used here.

6. **Euclidean or ℓ_2 norm:**

$$\| x \|_2 = \sqrt{\sum_{n=0}^{L-1}|x[n]|^2} = \sqrt{\mathcal{E}_x}.$$

This is the vector length in $\mathbb{R}^L$. norm(x,2).

Useful identities: $\mathcal{E}_x = L\mathcal{P}_x$, $x_{\text{RMS}} = \| x \|_2/\sqrt{L}$, and $\mathcal{P}_x = \sigma_x^2 + |\mu_x|^2$. Thus, **RMS** and standard deviation agree only when the mean is zero.

Uniform mid-rise quantization

Define the observed extrema and dynamic range:

$$X_{\min} = \min_n\{x[n]\}, \qquad X_{\max} = \max_n\{x[n]\}, \qquad R = X_{\max} - X_{\min}.$$

For $R > 0$, a uniform mid-rise quantizer with $N = 2^B$ cells (strips) has step size

$$\boxed{\Delta_N = \frac{R}{N}.}$$

The kth cell is $[X_{\min} + k\Delta_N, X_{\min} + (k+1)\Delta_N)$ for $k = 0, \ldots, N-2$. The last cell *also includes* $X_{\max}$. Its reconstruction amplitude is

$$r_k = X_{\min} + (k + 1/2)\Delta_N, \qquad k = 0, \ldots, N-1.$$

The clipped quantization index and quantized signal are

$$\begin{aligned} k_N[n] &= \min\left\{N-1, \max\left\{0, \left\lfloor \frac{x[n]-X_{\min}}{\Delta_N} \right\rfloor\right\}\right\}, \\ x_{q,N}[n] &= Q_N(x[n]) = X_{\min} + (k_N[n] + 1/2)\Delta_N. \end{aligned}$$

In particular, $k_N(X_{\max}) = N-1$ and $Q_N(X_{\max}) = X_{\max} - \Delta_N/2$. The maximum input *belongs to the last cell*; it is not itself a reconstruction value. Likewise $Q_N(X_{\min}) = X_{\min} + \Delta_N/2$. If $R = 0$, define $x_{q,N}[n] = x[n]$ for all n (a constant-signal special case).

Quantization error and distortion

Define the samplewise quantization error (original minus reconstructed) as

$$\boxed{e_N[n] = x[n] - x_{q,N}[n].}$$

The **total squared error** and **mean squared error (MSE)** are

$$\mathcal{E}_{e,N} = \sum_{n=0}^{L-1} |e_N[n]|^2 = \| x - x_{q,N} \|_2^2,$$

$$\boxed{\mathrm{MSE}(N)} = \boxed{\frac{1}{L}\sum_{n=0}^{L-1} \left|x[n] - x_{q,N}[n]\right|^2 = \frac{\mathcal{E}_{e,N}}{L}}.$$

MSE is the *average power of the error sequence*, not generally the variance of the error: $\mathrm{MSE}(N) = \sigma_{e,N}^2 + \left|\mu_{e,N}\right|^2$. The **root mean squared error** is $\mathrm{RMSE}(N) = \sqrt{\mathrm{MSE}(N)}$. For in-range samples of this mid-rise quantizer, $|e_N[n]| \le \Delta_N/2$; consequently $\mathrm{MSE}(N) \le \Delta_N^2/4$. These are deterministic bounds and do not require a stochastic noise model.

Useful additional performance definitions

Signal-to-quantization-noise ratio (SQNR): for $\mathrm{MSE}(N) > 0$,

$$\mathrm{SQNR}_{\mathrm{dB}}(N) = 10\log_{10}\left(\frac{\mathcal{P}_x}{\mathrm{MSE}(N)}\right).$$

This uses the original signal's mean-square power as the numerator; it is a measurement convention, not an assertion that error and signal are uncorrelated. If MSE is zero, SQNR is infinite for a nonzero signal.

High-resolution quantization-noise approximation: when errors are approximately uniform on $[-\Delta_N/2, \Delta_N/2]$ with negligible bias,

$$\mathrm{MSE}(N) \approx \frac{\Delta_N^2}{12} = \frac{R^2}{12N^2} = \frac{R^2}{12 \cdot 2^{2B}}.$$

Hence doubling N predicts one-quarter the MSE, and increasing B by one predicts a $10\log_{10}(4) \approx 6.02$ dB improvement in SQNR when signal power is fixed. This is an *approximation*, not an exact identity for deterministic audio or coarse quantization.

Peak absolute error: $e_{\max,N} = \max_n |e_N[n]|$. **Normalized MSE:** $\mathrm{NMSE}(N) = \mathrm{MSE}(N)/\mathcal{P}_x$ for $\mathcal{P}_x > 0$.

A **uniformly time-sampled** and **uniformly quantized signal** is considered in this assignment to be an analog signal if $\boldsymbol{20\,log_{10}\,\Delta_N < -90\ dB}$, where $\boldsymbol{\Delta_N}$ is defined as its **amplitude resolution**.

The audio file name is dtmfad.wav. The script uses the first **2205** samples, computes the quantizer range from those same samples, and computes metrics independently for each sequence.

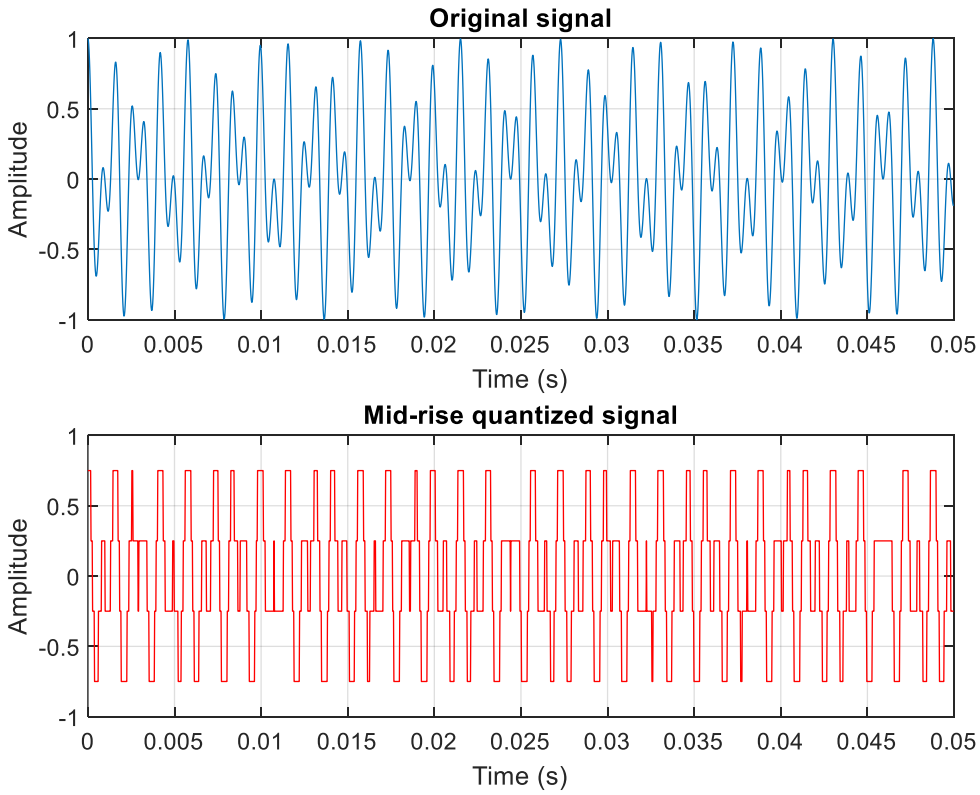
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REFERENCE: Julius O. Smith III, *Mathematics of the Discrete Fourier Transform (DFT), with Audio Applications*, 2nd ed., Center for Computer Research in Music and Acoustics, Stanford University. https://www.dsprelated.com/freebooks/mdft/.
https://www.dsprelated.com/freebooks/mdft/Signal_Metrics.html.

DESIGN PROJECT:

For the successful completion of this assignment, each member in each group must study, run, and modify the provided m-script in the zipped link.

01.- Run the **cuantizador_midrise.m** for a total of $\boldsymbol{L = 2205}$ samples to be processed ($\boldsymbol{50\ mSec}$), at $\boldsymbol{N = 4}$ quantization levels. **Proceed** to compute the **MSE** for $\boldsymbol{N = 4}$. See graph below for the output of the quantizer.



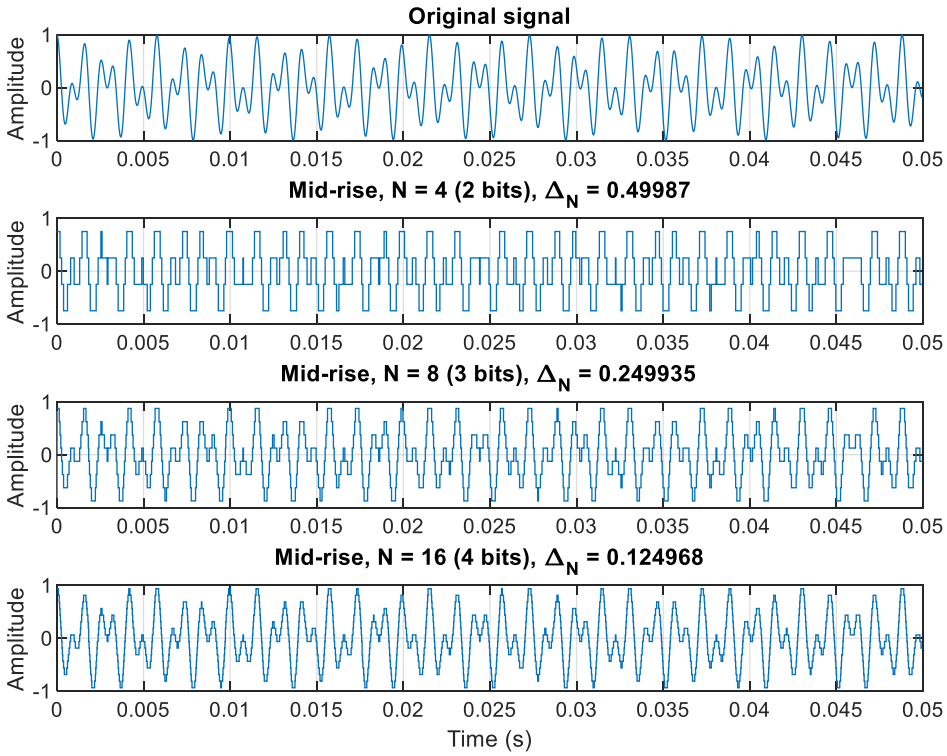
02.- Use your **DTFM** signal from **MCA02** to compute the six metrics for a fixed (**one per student**) $\boldsymbol{N \neq 4}$., for the original and quantized signals. Then, compute the **MSE** for the six quantization values:

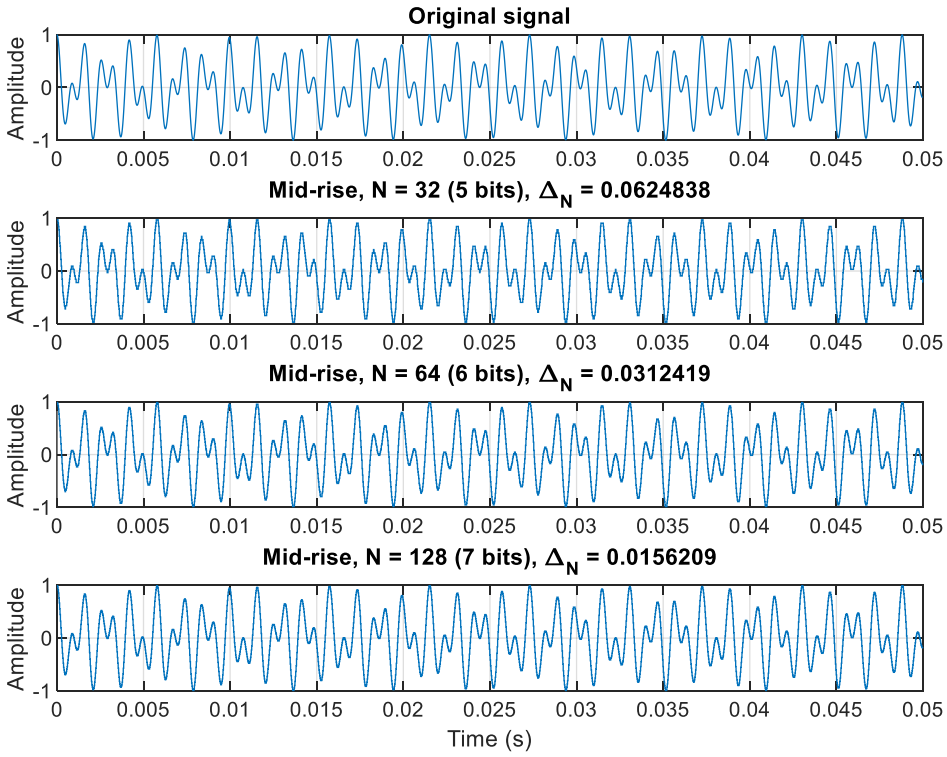
$$\boldsymbol{N = 4, 8, 16, 32, 64, 128}$$

04.- Plot a graph as the one above for your selected $\boldsymbol{N}$ value ($\boldsymbol{N}$ different for every student). Learn how and proceed to generate a **CSV** file as the one provided. Each student must generate her/his own **.csv** file.

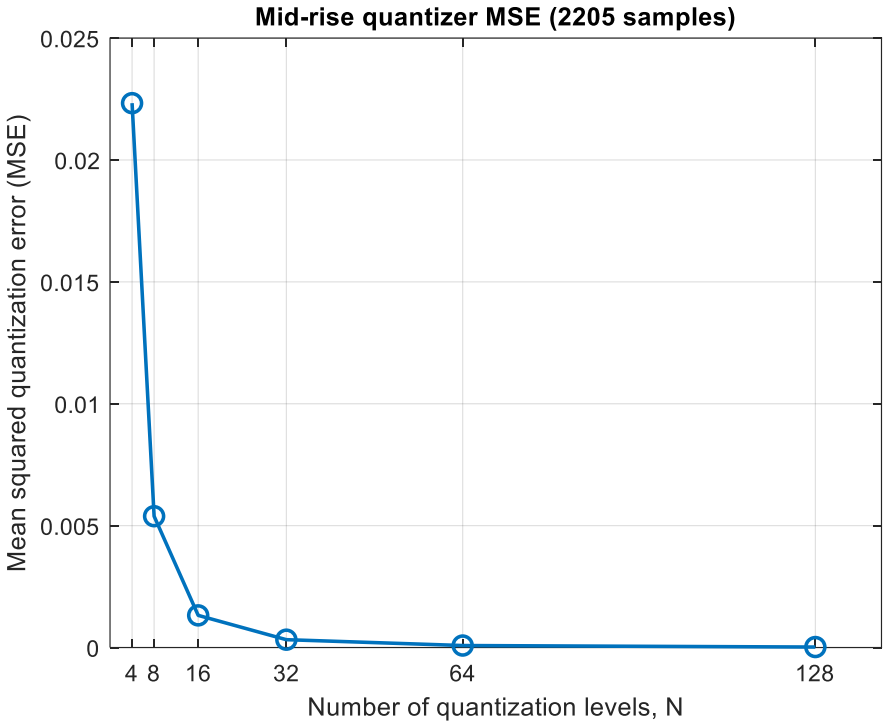
05.- Provide a separate plot for the **MSE** as the one given below.

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TAKE-HOME ASSIGNMENT GUIDELINES:

A.- This take-home work does not require a written result. It requires for each student group to send a **.zip** folder with the following items provided in the list below. There should be **NOTHING else** inside the requested **.zip** folder:

i.- Original, INEL4301_SYYY_GPZZ_MCA03_261009.pdf
ii.- Original, unaltered, **dtmfad.wav** and your selected (MCA02) **dtmf**
iii.- Original, unaltered, **inel4301_syyy_gpzz_sn00_tk00_mca03.m**
iv.- One mod. **inel4301_syyy_gpzz_sn00_tk00_mca03.m** per student.
v.- One **inel4301_syyy_gpzz_sn00_tk00_N000_mca03.csv** per student.

Refers to the class section number: **s000**
Refers to the student's group number: **gp00**
Refers to the student's task in the group: **tk00**
Refers to the student's number in the group: **sn00**
Refers to the student's signal quantization value: **N000**

B.- Name of **.zip** folder and the name of the **e-mail subject**:

INEL4301_SXXX_GPYY_MCA03

C.- Send e-mail, with the correct **e-mail subject**, and **.zip** file, with appropriate name, to the following e-mail address:

domingo.rodriguez1@upr.edu

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APPENDIX

Quantization Step Size Analysis: A 16-Bit Uniform Quantizer

This appendix details the analytical derivation used to find the delta (Δ) or quantization step size of a 16-bit uniform quantizer given a full-scale range from a minimum of -1 to a maximum of +1.

1. System Parameters

The fundamental variables provided for this configuration are as follows:

- **Number of bits (n):** 16
- **Maximum amplitude (Vmax):** +1
- **Minimum amplitude (Vmin):** -1

2. Signal Range Calculation

To determine the total size of the input domain, we compute the peak-to-peak dynamic range (Vpp):

Vpp = Vmax - Vmin

Vpp = 1 - (-1) = 2

3. Quantization Levels

The total number of unique binary representation levels (L) achievable with a 16-bit binary string is calculated using base-2 exponentiation:

L = 2^n

L = 2^16 = 65,536 levels

4. Step Size (Δ) Conventions

Depending on the exact physical or algorithmic design mapping of the boundary thresholds (such as mid-rise or mid-tread logic), signal processing typically utilizes one of two standard mathematical conventions to find the step size:

Convention A: Full-Interval Division (Standard Digital/ADC Framework)

In standard ADC structures, the full dynamic range is split symmetrically into equal intervals matching the maximum state space count:

Δ = (Vmax - Vmin) / 2^n

Δ = 2 / 65,536 = 1 / 32,768

Δ ≈ 3.0517578 × 10^-5

Convention B: Level-to-Level Spacing (Exact Edge Reconstruction)
If the highest and lowest voltage limits are locked exactly to the extreme reconstruction outputs, the range spans L - 1 divisions:
Δ = (Vmax - Vmin) / (2^n - 1)
Δ = 2 / 65,535
Δ ≈ 3.0518044 × 10^-5

Summary Table:

Convention Type	Formula	Exact Step Size (Δ)
Convention A (Full Range)	2 / 65,536	3.0517578 × 10^-5
Convention B (Bounded Levels)	2 / 65,535	3.0518044 × 10^-5

5. Normalized Signal to Noise Ratio (SNR) in dBs

Since the maximum value of the input signal $x(t)$ is assumed to be $+1\,V$, the signal $x(t)$ to noise $n(t)$ ratio (SNR) in decibels is given by:

$$SNR(dB) = 20Log_{10}\left\{\frac{x(t)}{n(t)}\right\} = 20Log_{10}\left\{\frac{1}{n(t)}\right\} = -20Log_{10}\{n(t)\}$$

We recall, from above, that $\boldsymbol{\Delta_N}$ was defined as the **amplitude resolution** for the mid-rise uniform quantizer (Standard Digital/ ADC).
If we assume **zero mean** additive white Gaussian noise (**AWGN**) and we also assume that most of the noise stays inside a single cell $\boldsymbol{\Delta_N}$, we have:

$$SNR(dB) = -20Log_{10}\left\{\frac{\boldsymbol{\Delta_N}}{\mathbf{2}}\right\} = -20Log_{10}\left\{\left(\frac{1}{2}\right)\left(\frac{2}{2^N}\right)\right\} = -20Log_{10}\{(2^{-N})\}$$

Thus, we have the following formulation for voltage-based normalized SNR, assuming that $\boldsymbol{M}$ is the number of bits used in the linear coder.

$$\boldsymbol{SNR(dB) = -20Log_{10}\{(2^{-N})\} = 20NLog_{10}\{2\} = 6.0206N;\quad N = 2^M}$$